



Q1) Given the following data for this reaction: $A+B+C \rightarrow D$

Experiment	[A]	[B]	[C]	RATE (M/s)
1	0.10 M	0.20 M	0.30 M	0.008
2	0.30 M	0.20 M	0.30 M	0.216
3	0.10 M	0.20 M	0.50 M	0.0133
4	0.10 M	0.30 M	0.50 M	0.020

The rate law expression for the reaction is:

A. Rate = $k[A]^1[B]^1[C]^3$

D. Rate = $k[A]^3[B]^1[C]^2$

B. Rate = $k[A]^1[B]^3[C]^1$

E. Rate = $k[A]^3[B]^1[C]^1$

C. Rate = $k[A]^2[B]^1[C]^2$

Q2) What is the activation energy (in kJ) of a reaction whose rate constant increases by a factor of 35 upon increasing the temperature from 300 K to 360 K?

A. 27

B. 35

C. 42

D. 53

E. 69

Q3) The half-life for a first-order reaction ($A \rightarrow B$) is 112.5 s. What was the original concentration if, after 2.0 minutes, the reactant concentration is 0.062 M?

A. 0.83 M

B. 0.069 M

C. 0.091 M

D. 0.075 M

E. 0.13 M

Q4) The half-life for the second order reaction: $A \rightarrow B$ is 1570 seconds. What fraction of an initial amount of A remains after 23549 seconds if the initial concentration of A is 0.55 M?

A. 0.333

B. 0.167

C. 0.051

D. 0.0625

E. 0.0313

Q5) How many times will the rate of the reaction ($3X + Y \rightarrow X_2Y$) change when the concentration of the substance Y is doubled and that of X is constant? (Hint: the rate law is Rate (r) = $k[X]^3[Y]$)

A. $r_2 = 24r_1$

B. $r_2 = 5r_1$

C. $r_2 = 2r_1$

D. $r_2 = 4r_1$

E. none

Q6) For the reaction: $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ If ($\Delta S^\circ_{rxn} = -177 J K^{-1} \cdot mol^{-1}$, $\Delta G^\circ_{rxn} = -120.7 kJ/mol$). Calculate the enthalpy of reaction in kJ per mole NO.

A. -43.3

B. -86.7

C. 43.3

D. 86.7

E. 173.4

Q7) Using the standard free energies of formation ($kJ mol^{-1}$)

($NO_2(g)$, $\Delta G^\circ_f = +51.84$) ($NO(g)$, $\Delta G^\circ_f = +86.69$) ($SO_2(g)$, $\Delta G^\circ_f = -300.0$) ($SO_3(g)$, $\Delta G^\circ_f = -370.0$)

Calculate the standard free energy change, ΔG° , for the reaction: $NO_2(g) + SO_2(g) \rightarrow NO(g) + SO_3(g)$

A. -35.15 kJ

B. -104.9 kJ

C. -429.2 kJ

D. -619.6 kJ

E. -808.5 kJ

Q8) The reaction, $A + B \rightarrow C$ is spontaneous at low temperatures but non-spontaneous at high temperatures. we can deduce that the values of ΔH° and ΔS° are constant over this temperature range then:

A. $\Delta H < 0$ and $\Delta S > 0$

B. $\Delta H > 0$ and $\Delta S < 0$

C. $\Delta H < 0$ and $\Delta S < 0$ ✓

D. $\Delta H > 0$ and $\Delta S > 0$ ✓

E. $\Delta G = 0$ at 298 K

Q9) The Third Law of Thermodynamics states:

A. $\Delta E_{sys} = q + w$

B. The entropy of the universe increases for spontaneous processes.

- C. $\Delta S = 0$ for a pure element in its most stable state at standard conditions.
 D. $S = 0$ for a perfect crystal of a pure substance at 0 K.
 E. none of the above

Q10) Which one of the following reactions has $\Delta S > 0$?

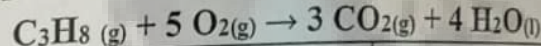
- A. $C_{12}H_{20}(l) + 17 O_2(g) \rightarrow 12 CO_2(g) + 10 H_2O(l)$ -
 B. $NH_4Cl(s) \rightarrow NH_3(g) + HCl(g)$ +
 C. $2 C_2H_2(g) + 5 O_2(g) \rightarrow 4 CO_2(g) + 2 H_2O(s)$ -
 D. $Ba(OH)_2(s) + 2 HCl(g) \rightarrow BaCl_2 \cdot 2H_2O(s)$ -
 E. $(CH_3)_2CO(l) + 4 O_2(g) \rightarrow 3 CO_2(g) + 3 H_2O(l)$ -

Q11) The value of ΔH° for the reaction below is -72 kJ. Calculate amount of heat released when 242.7 grams of HBr (M.w=80.9g/mol) is formed in this reaction. $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$
 A. -144 B. -72 C. -108 D. -36 E. -72

Q12) A 1.5L of an ideal gas is expanded isothermally against external pressure equal to 2.96 atm at 25°C. During expansion, the gas absorbed 549 joule of heat. What is the volume of the gas after expansion?
 A. 0.83 L B. 2.33 L C. 0.667 L D. 8.33 L E. 3.33 L

Q13) A correct description for the electron configuration of a vanadium atom (^{24}Cr) is
 A. $[Ar] 4s^1 3d^5$ B. $[Ar] 4s^2 3d^3$ C. $[Ar] 4s^2 3d^5$ D. $[Ar] 3d^5$ E. $[Ar] 3s^2 3d^4$

Q14) Based on the information bellow, which expression represents the enthalpy for the following reaction:



Reaction	Enthalpy of reaction
$C_3H_8(g) \rightarrow 3C(s) + 4 H_2(g)$	X
$CO_2(g) \rightarrow C(s) + O_2(g)$	Y -3
$4H_2(g) + 2 O_2(g) \rightarrow 4H_2O(l)$	Z -3

- A. $-X + 3Y + 2Z$ B. $X - 3Y + Z$ C. $-2X - 3Y + 2Z$ D. $-X - 3Y + 2Z$ E. $-X + 3Y - 2Z$

Q15) The combustion of titanium with oxygen produces titanium dioxide: $Ti(s) + O_2(g) \rightarrow TiO_2(s)$
 When 2.79 g of titanium (47.9g/mol) is combusted in a bomb calorimeter, the temperature of the calorimeter increases from 25.00 °C to 41.60 °C. If the heat capacity of the calorimeter equal 9.84 kJ/K. Calculate enthalpy of reaction (in KJ) for the combustion of a mole of Ti in this calorimeter.
 A. -163.3 B. -165.8 C. -326.6 D. -3800 E. -3798

Q16) Which atom has the smallest first ionization energy?
 A. ^{13}Al B. ^{19}K C. ^{31}Ga D. ^{12}Mg E. ^{11}Na

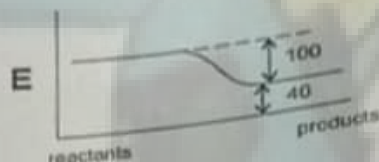
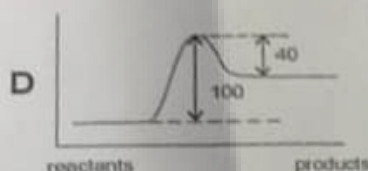
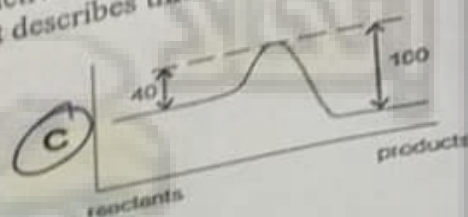
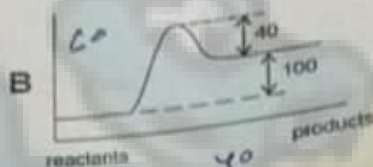
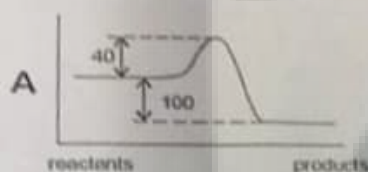
Q17) Which one of the following statements is false?
 A. Internal energy is a state function.
 B. Enthalpy is an intensive property.
 C. Enthalpy is a state function.
 D. The enthalpy change for a reaction depends on the state of the reactants and products.
 E. The enthalpy of a reaction is equal to the heat of the reaction.



Q18) What is the mass % of hydrogen in dimethylsulfoxide (C_2H_6SO)? (At. wt g/mol: C=12, H=1, S=32, O=16)
 A. 60.0 B. 20.5 C. 30.7 D. 7.69 **E. 9.8**

Q19) Under appropriate conditions, nitrogen and hydrogen undergo a combination reaction to yield ammonia: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$. The reaction percentage yield = 61.5%. Calculate the actual yield of NH_3 produced if 7.1 g sample of N_2 are allowed to react with enough amount of H_2 . (At. wts: N=14, H=1 g/mol)
 A. 5.3 B. 6.7 C. 4.6 D. 1.5 **E. 17.4**

Q20) An endothermic reaction ($X \leftrightarrow Y$) has enthalpy equal to 60 kJ, and activation energy equal to 40 kJ for its reverse reaction. Which of one of the energy diagrams shown below best describes this reaction.



Q21) A 25.00 mL of a vinegar solution is diluted to 250.0 mL. If the concentration of the diluted vinegar solution is 0.05287 M. What is the concentration of the original vinegar?
 A. 0.2578 M **B. 0.5287 M** C. 0.7825 M D. 0.8527 M E. 0.9978 M

Q22) If 50.0 mL of water is added to 50.0 mL of 0.60 M solution of $AlCl_3$. What is the final concentration of chloride ion (Cl^-) after adding water?
 A. 1.8 M **B. 0.3M** C. 0.2M D. 0.6M E. 0.9 M

Q23) What is the molar concentration of a solution prepared by dissolving 10.0 g of cobalt(II) chloride dihydrate ($CoCl_2 \cdot 2H_2O$ M.w=165.87 g/mol) in enough amount of ethanol to make a 114.6 mL solution.
 A. 0.121 M B. 0.263M C. 0.526 M D. 0.734 M E. 0.972 M

Q24) For the reaction: $Na_2CO_3(s) + 2HNO_3(l) \rightarrow 2NaNO_3(s) + CO_2(g) + H_2O(l)$, how many grams of Na_2CO_3 (molar mass = 106.0 g/mol) are required to react completely with 152.2 mL of 0.155 M HNO_3 ?
 A. 0.125 g B. 0.205 g C. 0.410 g D. 1.250 g E. 2.050 g

Q25) A 20 g of common salt is dissolved in 400 g of water. Calculate the mass percent concentration of the solution (w/w%).
 A. 3.75 % B. 3.61 % **C. 4.76 %** D. 7.45 % E. 5.88%

Good luck